Business Problem & Key Findings

Business Problem & Prediction Target

An investment company plans to invest $10 million in Lending Club loans and needs to identify loans at higher risk of default. We analyze 7,151 loans issued in 2015, restricted to 36-month loans for credit card financing. The target variable is loan_status, encoded as 1 = Charged Off and 0 = Fully Paid. We develop and compare Logistic Regression and Decision Tree models using information available at the time of investment to support loan selection.

Target Distribution and Class Imbalance

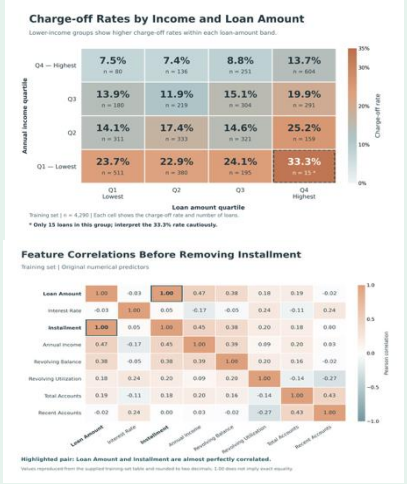
The dataset contains 83.1% Fully Paid and 16.9% Charged Off loans. Predicting every loan as Fully Paid would achieve 83.1% accuracy while missing every default. We therefore prioritize recall to reduce missed defaults, while monitoring precision to assess false alarms and ROC AUC to evaluate risk ranking. Both models use balanced class weights to give charged-off loans greater weight during training.

Exploratory Data Analysis

EDA was conducted on the **training set** following a stratified **60%** training **/ 20%** validation **/ 20%** test split.

- **Interest rate:** Higher interest rates were associated with higher charge-off rates. Interest rate had the strongest correlation with the target among the numerical features examined (**r = 0.25**).
- **Income and loan size:** Within each loan-amount band, the lowest-income group had higher charge-off rates than the highest-income group. This motivated exploring **loan-to-income ratio** as a candidate feature. The highest observed rate (**33.3%**) was based on only **15 loans**, requiring cautious interpretation.
- **Borrower characteristics:** Renters had a higher charge-off rate than mortgage holders (**21.6% vs. 12.8%**). Verified-income borrowers also had higher rates than unverified borrowers (**22.8% vs. 13.4%**), alongside a greater concentration of high-interest loans. These associations do not establish causation.
- **Skewed distributions:** Annual income and revolving balances were strongly right-skewed. Large values were examined rather than automatically removed, informing the use of log transformations for Logistic Regression.





Feature Selection & Preprocessing

- **Feature selection:** Removed id and member_id as identifiers, term and purpose as constant features, and total_pymnt to prevent target leakage.
- **Multicollinearity:** Removed installment because it was almost perfectly correlated with loan_amnt ($r \approx 1.00$), reducing redundancy and supporting clearer Logistic Regression coefficient interpretation.
- **Missing values and encoding:** Used training-set median imputation for missing numerical values and one-hot encoding for home_ownership and verification_status.
- **Model-specific preprocessing:** Applied log transformations to annual_inc and revol_bal, followed by numerical standardization for Logistic Regression. Decision Tree required no scaling. All preprocessing was fitted on training data only.

Columns	Treatment	Reason
id, member_id	Drop	Identifiers without meaningful credit-risk information.
term, purpose	Drop	Constant across this dataset; cannot distinguish outcomes.
total_pymnt	Drop	Payments received after issuance would cause target leakage.
installment	Drop	Nearly perfectly correlated with loan_amnt; remove redundancy to reduce multicollinearity.
annual_inc, revol_bal	Log + standardize (LR)	Reduce strong right skew and put features on a common scale.
loan_amnt, int_rate, revol_util, total_acc, acc_open_past_24mths	Standardize (LR)	Align numerical scales for regularized Logistic Regression.
home_ownership, verification_status	One-hot encode (both)	Represent categories without imposing an artificial order.
loan_to_income, revol_bal_to_income	Candidate features	Scale for LR if retained; use validation results to assess whether these ratios add value.

Source: Lending Club dataset via Kaggle (jayantmathur);

Model Results & Recommendation

Model Development

We trained a Logistic Regression model as an interpretable baseline and a Decision Tree to capture nonlinear relationships and interactions between features. Both models used balanced class weights to address class imbalance. We evaluated them on the same validation set using a classification threshold of 0.40.

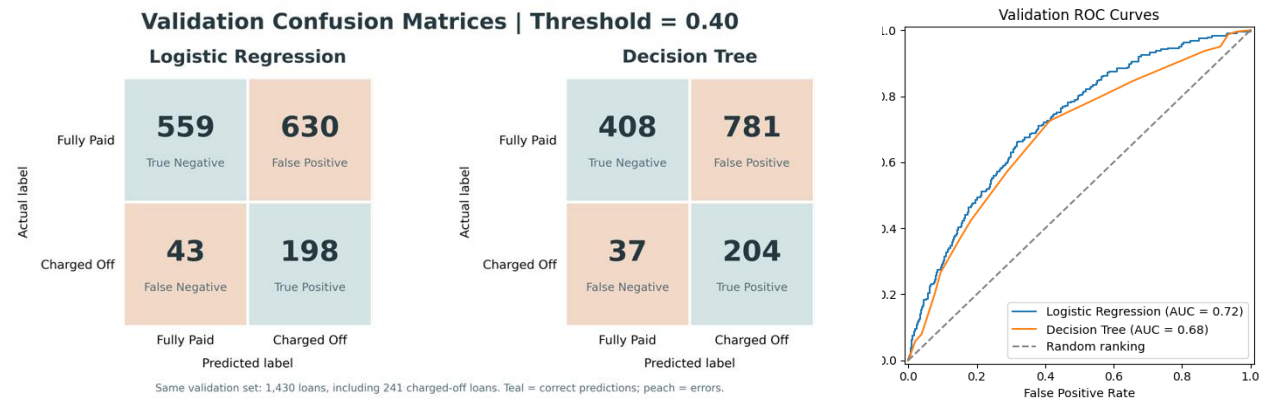
Validation Results & Interpretation

- Decision Tree achieved slightly higher recall, identifying **84.6% of charged-off loans**, compared with **82.2%** for Logistic Regression. However, Logistic Regression achieved higher precision, F1 and ROC AUC while flagging fewer loans for review. Its **23.9% precision** means that approximately one in four flagged loans actually charged off, so false alarms remain a substantial limitation.
- Lowering the Logistic Regression threshold from **0.50 to 0.40** increased recall from **66.4% to 82.2%**, but reduced precision from **28.9% to 23.9%**. This reflects the trade-off between detecting more defaults and flagging more loans that would repay.

Validation performance at threshold 0.40

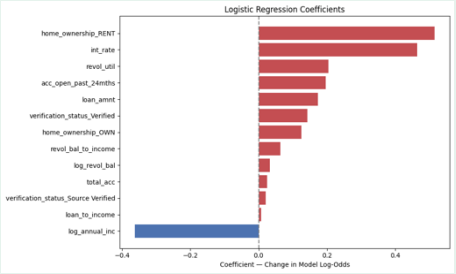
Model	Recall	Precision	F1	ROC AUC	Flagged
Logistic Regression	82.2%	23.9%	0.370	0.720	57.9%
Decision Tree	84.6%	20.7%	0.333	0.684	68.9%

- At a threshold of **0.40**, Logistic Regression identified **198 of 241 charged-off loans**, missed **43**, and incorrectly flagged **630 fully paid loans**. Decision Tree detected **204 defaults** and missed **37**, but incorrectly flagged **781 fully paid loans**. It therefore captured **6 additional defaults at the cost of 151 additional false positives**, supporting Logistic Regression as the preferred model when balancing default detection with review workload.

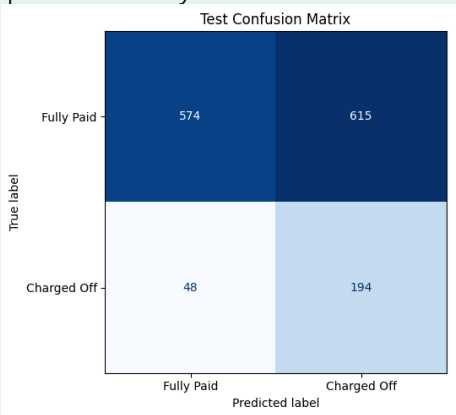


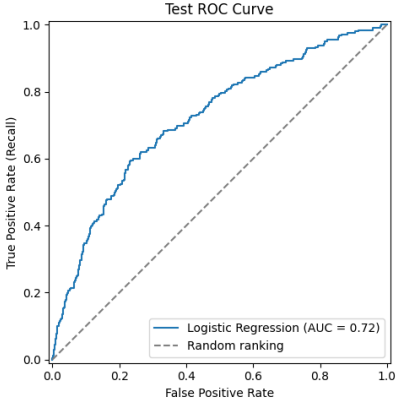
Key Risk Factors & Final Model Selection

- **Key Risk Factors:** Holding other features constant, renting rather than having a mortgage, higher interest rates, greater revolving utilization and more recently opened accounts were associated with higher charge-off risk. Higher income was associated with lower risk.



- **Final Model Selection and Rationale:** We selected **Logistic Regression at threshold 0.40** for its higher precision, F1 and ROC AUC, with fewer loans flagged for review. Test performance was similar to validation: **80.2% recall, 24.0% precision and 0.72 ROC AUC**. We recommend using the model to prioritize analyst review.





Source: Lending Club dataset via Kaggle (jayantmathur);

Business Application & Deployment

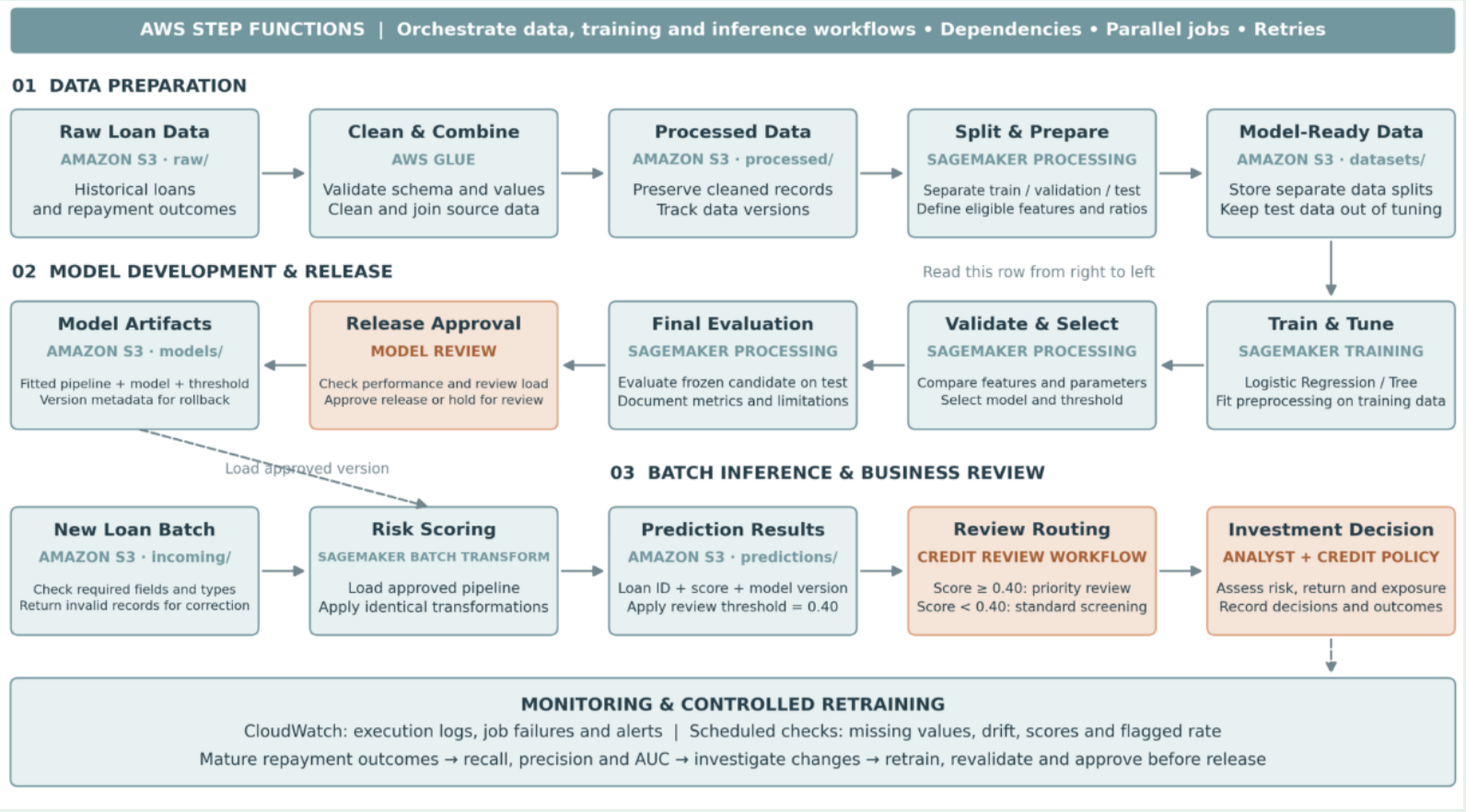
Risk Scoring in the Loan Review Process

The model would score eligible loans using borrower and loan information available at the time of investment. Higher-scoring loans would receive priority for analyst review, supporting loan selection for the proposed $10 million investment. Final decisions would also consider credit policy, expected returns and portfolio exposure.

Risk Threshold & Human Review

Loans with scores ⩾ 0.40 would be flagged for additional review; loans below the threshold would continue through standard screening. Analysts would assess flagged loans before making investment decisions. A flag would not automatically trigger rejection, and a lower score would not guarantee repayment. Because the model uses balanced class weights, scores should be treated as risk-ranking signals, rather than calibrated default probabilities.

Proposed AWS Deployment Workflow



Business Impact, Monitoring & Next Steps

- **Business value & review capacity:** The model supports consistent risk screening and prioritizes analyst review. At threshold **0.40, 57.9% of validation loans** were flagged, so the operating threshold should reflect review capacity and missed-default costs. Financial benefits require further assessment.
- **Monitoring & rollout:** Begin with a limited batch-scoring pilot. Monitor job failures, missing values, feature drift and flagged rates. Once repayment outcomes mature, track **recall, precision and ROC AUC**. Validate retrained models before release and retain previous versions for rollback.
- **Data limitations & feature engineering:** The limited set of available attributes restricts feature engineering opportunities. We created **loan-to-income and revolving-balance-to-income ratios**; richer credit history and debt information available at investment time could support additional predictors.
- **Further optimization:** Validate on **newer loan cohorts**, assess probability calibration and refine the threshold. Compare engineered features against the baseline and explore **Random Forest or XGBoost**, balancing validated performance gains with interpretability and operational complexity.

Source: Lending Club dataset via Kaggle (jayantmathur);